Literature Review

Organ Transplant Donor–Recipient Matching

Supporting Document for PBL Project

Smart Organ Transplant Matching System

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This document reviews three research papers on organ donor–recipient matching, summarizes what each paper did, identifies gaps left open across the literature, and explains how the proposed Smart Organ Transplant Matching System — a DSA-based project using Trees, Heaps, and Graphs — is positioned differently from the approaches used in these papers.

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1. Purpose of This Document

This document reviews the three research papers cited in the Month-1 PBL Progress Report, summarizes what each paper actually did, identifies the gaps left open by each, and explains how the proposed **Smart Organ Transplant Matching System** (a DSA-based project using Trees, Heaps, and Graphs) is positioned differently from — and is a deliberate simplification of — the approaches used in these papers.

The three papers reviewed are:

1. **Syed et al. (2024)** — Potential of New 250-Nautical Mile Concentric Circle Allocation System for Improving the Donor/Recipient HLA Matching: Development of New Matching Algorithm. *Transplant Immunology* 87, 102146.
2. **Majouni et al. (2025)** — A Heterogeneous Bipartite Graph Framework for Donor-Recipient Matching in Kidney Transplantation.
3. **Melehy et al. (2026)** — Liver Transplant Donor-Recipient Matching with Offline Reinforcement Learning. *npj Digital Medicine* 9:351.

2. Paper-by-Paper Review

2.1 Syed et al. (2024) — 250-Nautical Mile HLA Allocation Simulation

What the paper is about: The paper studies whether the new US kidney allocation policy (the 250-nautical-mile concentric-circle rule introduced in 2018) can be used to find immunologically better-matched kidney donors for patients, instead of just any ABO-compatible donor.

What they actually did:

- Used real SRTR (Scientific Registry of Transplant Recipients) data for 672 waitlisted patients and 6,558 deceased donors around University of Toledo Medical Center (UTMC), 2014–2019.
- Built an in-house simulation/allocation algorithm that imputes high-resolution HLA types from low-resolution data using haplotype frequency tables.
- Calculates a **Hydrophobic Mismatch Score (HMS)** — a continuous, amino-acid-level immunogenicity score — for every possible donor–recipient pair.
- Enforces ABO blood-type compatibility as a hard constraint, and optionally KDPI ≤ 50% (Kidney Donor Profile Index, a donor-quality score).
- Picks the lowest-HMS (least immunogenic) available donor for each recipient, removing that donor from the pool once used — a greedy one-to-one assignment.
- Compared simulated matches against a real historical cohort (2000–2010) using Kaplan–Meier survival analysis.

Key results:

- 99.8% of patients could be matched with a low-immunogenicity (HMS ≤ 10), KDPI ≤ 50% donor.
- Simulated matches had far lower average HMS (1.4) than real-world transplants (6.3).
- Projected median graft survival rose from 13.4 years (real cohort) to 18.2 years (simulated) — a gain of ~4.9 years per transplant.
- Each patient typically had access to 100+ potential well-matched donors within the 250-mile radius (median ~141–229 donors per year).

How it works, structurally: A greedy bipartite assignment algorithm — for each recipient, search all eligible donors, filter by hard constraints (ABO, KDPI), rank by a continuous immunogenicity score (HMS), and assign the best available one. No machine learning is used; it is a rules + scoring + greedy-selection simulation.

2.2 Majouni et al. (2025) — Heterogeneous Bipartite Graph + GNN Framework

What the paper is about: A methodology/architecture paper (not yet a fully evaluated system) proposing a graph neural network (GNN) approach to donor–recipient matching for kidney transplants, framed as a "feature set matching" problem.

What they actually did:

- Modeled donors and recipients as the two node sets of a heterogeneous bipartite graph built from SRTR data (5,000 donor nodes, 5,000 recipient nodes; donors have 26 features).
- **Compatibility edges** — connect donor–recipient pairs that transplanted and survived past a threshold (5 or 8 years). Sparse, one-to-one.
- **Knowledge-driven edges** — derived from the Baskin-Bey Recipient Risk Score (RRS) and a Deceased Donor Score (DDS, 0–39 points; age, hypertension, estimated GFR, HLA mismatch, cause of death). Donors graded A–D, matched to recipient risk groups RG1–RG4.
- **Blood-type and cluster-based edges** — encode ABO compatibility rules and cluster donors/recipients with historically higher match-success likelihood.
- **Time-based edges** — capture waitlist urgency, organ viability windows, and disease progression.
- The resulting graph has ~18 million edges connecting 5,000 donors to 5,000 recipients.
- Proposed pipeline: embed the heterogeneous graph → pass through a GNN → project donors and recipients into a shared embedding space → compute a compatibility score via dot product → output P(Donor|Recipient).

Key results: This paper is largely conceptual/architectural — it presents the graph construction and GNN pipeline design in detail but does **not report trained-model performance metrics** (no AUROC, accuracy, or survival outcomes). GNN training and evaluation are positioned as future work.

How it works, structurally: Multi-relational graph representation learning. Instead of one scalar score (like HMS), it encodes several kinds of relationships as different edge types in the same graph, then lets a GNN learn a compatibility embedding rather than relying on a single hand-engineered formula.

2.3 Melehy et al. (2026) — Liver Transplant Matching with Offline Reinforcement Learning

What the paper is about: Unlike the other two papers (essentially "given a donor, pick the best static match"), this paper reframes donor-recipient matching for liver transplantation as a sequential decision-making problem over time: at every point on the waitlist, should the clinician wait, delist, or transplant the patient with an available organ?

What they actually did:

- Used SRTR data: 45,635 liver-transplant candidates (2017–2022), converted into Markov Decision Process (MDP) episodes — each patient's MELD score (severity-of-liver-disease score) over time forms a trajectory of states.
- At each state (time step), the possible actions are: **Wait**, **Delist**, or **Transplant** (with a specific available donor).

- Designed a reward function combining short-term reward/penalty from MELD change, and long-term reward/penalty for terminal outcomes (graft failure, successful transplant, waitlist death, recovery).
- Trained an offline Reinforcement Learning model using **Conservative Q-Learning (CQL)** — a Double DQN variant with a conservatism regularizer, since the model must learn from historical data without live experimentation.
- Compared against baselines: simple MELD-threshold rules (e.g., “transplant whenever MELD ≥ 15”), and static supervised-learning predictive models (logistic regression, XGBoost).

Key results:

- The RL model avoided **73%** of donor-recipient pairs that led to real-world graft failure or death (GFPR), while preserving **93%** of real successful transplants (GSPR).
- It identified alternative donors for **47%** of patients who, in reality, died waiting.
- It outperformed both naive MELD-threshold rules and static ML predictive models on these clinically meaningful metrics simultaneously.

How it works, structurally: A temporal, policy-learning approach — it doesn't just score a single donor-recipient pair; it learns when to act across a whole patient trajectory, balancing competing risks (death from waiting vs. death from a bad transplant) over time.

3. Side-by-Side Comparison

Aspect	Syed et al. (2024)	Majouni et al. (2025)	Melehy et al. (2026)
Organ	Kidney	Kidney	Liver
Core technique	Rule-based greedy simulation + HMS score	Heterogeneous bipartite graph + GNN (proposed)	Offline RL (CQL / MDP)
Data source	SRTR	SRTR	SRTR
Matching philosophy	Best static match at allocation time	Learned similarity/compatibility embedding	Sequential wait/delist/transplant decision
Time dimension?	No (single-point)	Partially (time-edges as concept)	Yes — core to the model
Validated with outcomes?	Yes (graft survival gain, p<0.001)	No (architecture only)	Yes (GFPR, GSPR, CIR, CRR)
Computational complexity	Low–moderate	Very high (18M-edge GNN)	Very high (deep RL, 214K+ states)
Real-time use?	No — batch simulation	No — research prototype	No — “not intended for operational use”
Needs HLA typing?	Yes	Partially (one DDS input)	No (uses MELD, not HLA)

Table 1: Comparison of core technique, data handling, and validation across the three papers.

4. Gaps Identified Across the Three Papers

After reviewing all three papers together, the following gaps stand out — these gaps are the actual justification for a DSA-first project like the Smart Organ Transplant Matching System:

Gap 1 — Heavy, research-grade computation.

All three approaches are computationally heavy, data-hungry, and research-grade — not deployable as a lightweight, transparent decision-support tool. Syed et al. need high-resolution HLA imputation; Majouni et al. need a GNN trained on an 18-million-edge graph; Melehy et al. need to train a Q-network over 1.25 million iterations. None of these are things a hospital's waitlist software could run instantly on a laptop or explain to a coordinator in real time.

Gap 2 — No dynamic waitlist handling.

None of the three focuses on the data-structure problem of maintaining a live, constantly changing waiting list. All three treat the waiting list as a static batch dataset (a fixed SRTR snapshot) processed once, offline. In reality, patients are added, removed, or re-prioritized continuously. None of the papers discuss how to efficiently insert, delete, search, or re-rank patients in real time — this is precisely a classic tree/heap data-structures problem.

Gap 3 — Weak explainability.

Explainability and traceability are weak, especially in the ML/GNN and RL approaches. Majouni et al.'s GNN produces a similarity score from a learned embedding space that is hard for a clinician to interpret. Melehy et al.'s CQL model is a black-box Q-function. Only Syed et al.'s HMS-based approach is reasonably transparent, but it still requires specialized immunogenetics expertise to interpret.

Gap 4 — Inconsistent handling of geography/urgency.

Geographic/paired-exchange chains and urgency-based prioritization are handled inconsistently. Syed et al. handle geography (250-mile radius) but not urgency ranking. Majouni et al. mention time-based/urgency edges only conceptually, without an evaluated implementation. Melehy et al. handle severity (MELD) over time but not multi-donor chain relationships (e.g., paired kidney exchange).

Gap 5 — No unified, lightweight structure.

No paper combines “fast search over a large candidate pool” with “graph-based compatibility” with “urgency-based prioritization” in one lightweight, unified structure. Each paper solves one slice of the problem using a heavyweight technique, but none integrates ordered search + priority ranking + relationship-graph traversal into a single efficient system.

5. How the Proposed Solution Differs

The current project deliberately does not try to replicate the statistical/ML sophistication of these papers (HLA immunogenicity modeling, GNN embeddings, or offline RL policies are out of scope for a DSA-II course project). Instead, it targets the structural gap identified above: **efficient, explainable, real-time organization of the matching problem using classical data structures.**

Research Paper Technique	Proposed Project's Analogous Piece	Why It's Different
Greedy HMS-based donor search (Syed et al.)	Graph / adjacency list of donor–recipient compatibility (blood group, tissue type)	Same idea of searching compatible donors, but using a simple, explainable adjacency structure instead of amino-acid-level immunogenicity scoring — feasible to compute instantly without specialized lab data.
Multi-relational compatibility graph + GNN (Majouni et al.)	Graph representation of donor–recipient relationships, including potential paired-exchange chains	Keeps the idea of representing multiple compatibility relationships as a graph, but replaces the learned GNN embedding with direct graph traversal (BFS/DFS or adjacency lookups) — transparent and traceable, no training data or GPU required.
MELD-based sequential decisions over time (Melehy et al.)	Max-Heap based urgency prioritization of the recipient waiting list	Captures the same core need — who should be prioritized right now — using a heap's O(log n) insert/extract-max instead of a trained RL policy. Simpler to maintain, at the cost of not learning long-term optimal timing strategies.
Static batch allocation on a fixed dataset (all three)	AVL Tree for the waiting list, enabling live insert/delete/search as patients register, get transplanted, or are removed	Directly addresses Gap 2 — none of the three papers model the waitlist as a dynamic, continuously updated structure.
Black-box scoring (GNN, RL)	Clearly defined tree/heap/graph operations with visible intermediate state	Directly addresses Gap 3 (explainability) — every match/priority decision can be traced back to a specific comparison, traversal, or heap operation.

Table 2: Mapping of each paper's core technique to the proposed project's analogous DSA-based component.

In short, the reviewed papers optimize primarily for predictive/statistical accuracy using domain-specific medical scoring (HLA/HMS), learned graph embeddings (GNN), or learned temporal policies (RL) — all of which require large labeled datasets, domain expertise, and heavy computation, and none of which is intended for real-time list management (Melehy et al. explicitly state their model is “not intended for operational use in its current form”). The proposed **Smart Organ Transplant Matching System** instead targets the systems/data-structures layer that all three papers implicitly assume already exists — a fast, transparent, continuously maintainable way to store, search, rank, and connect donor and recipient records — using Trees (waitlist organization), Heaps (urgency prioritization), and Graphs (compatibility relationships), which are the Unit 1 and Unit 2 topics of the DSA-II course.

6. Conclusion

All three papers confirm that donor-recipient matching is a genuinely hard, high-stakes problem that current allocation systems handle imperfectly — whether the limitation is geographic-only matching without deep immunogenicity (implicit in the real-world baseline Syed et al. compare against), a lack of trained/evaluated learning models (Majouni et al.), or the acknowledged non-operational status of

state-of-the-art RL models (Melehy et al.). None of them, however, addresses the more foundational systems question of how to organize a constantly changing candidate/donor pool for fast search, ranking, and relationship-tracing — which is the specific gap this PBL project's tree-, heap-, and graph-based design is intended to explore at an educational/prototype level.

7. References

[1] Syed, F.J., Bekbolsynov, D., Green II, R.C., Kaur, D., Ekwenna, O., Sindhwani, P., Rees, M., & Stepkowski, S. (2024). Potential of new 250-nautical mile concentric circle allocation system for improving the donor/recipient HLA matching: Development of new matching algorithm. *Transplant Immunology*, 87, 102146. https://doi.org/10.1016/j.trim.2024.102146

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